UNIT 2 · 8 hours

Time Response of Zero and First Order Systems



Time Domain Analysis of Control Systems

Control Systems (22EEG53)
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Learning objectives

- Describe a zero-order system
- Derive the step, ramp and impulse response of a first-order system
- Use the time constant to find rise time and settling time
- Solve numericals on first-order systems

Syllabus context

Time domain analysis of control systems: standard test signals, time response of zero, first and second order systems, time domain specifications, type of systems, static error constants and steady-state error, generalized error series.

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Zero-order system

- Transfer function G(s) = K (a pure gain).
- Output follows the input instantly: c(t) = K r(t).
- Example: a potentiometer (output voltage ∝ wiper position), ideal amplifier.
- No dynamics: no transient, no poles.

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First-order system

G(s) = C/R = 1/(1 + sT)

standard form

Step: c(t) = 1 – e^(–t/T)

R = 1/s

Impulse: c(t) = (1/T) e^(–t/T)

R = 1

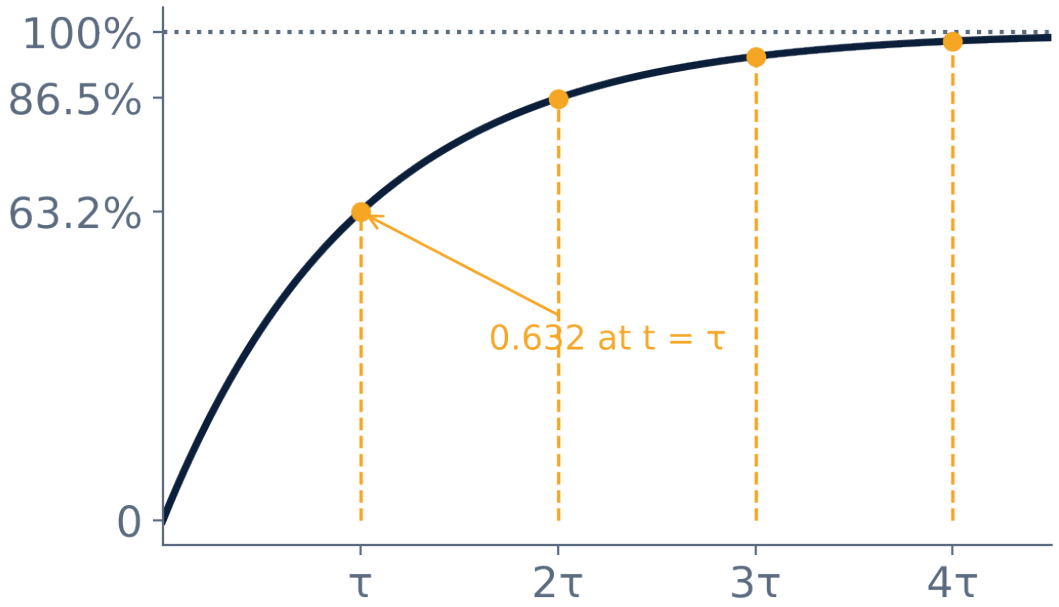
Ramp: c(t) = t – T + T e^(–t/T)

R = 1/s²

- T = time constant (s); single pole at s = −1/T.
- Examples: RC circuit (T = RC), thermometer, tank level.

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Unit step response



- At t = T: 63.2 % of final value.
- 2T: 86.5 %; 3T: 95 %; 4T: 98.2 %.
- Initial slope = 1/T: would reach final value in T if it continued.
- No overshoot: no oscillation.

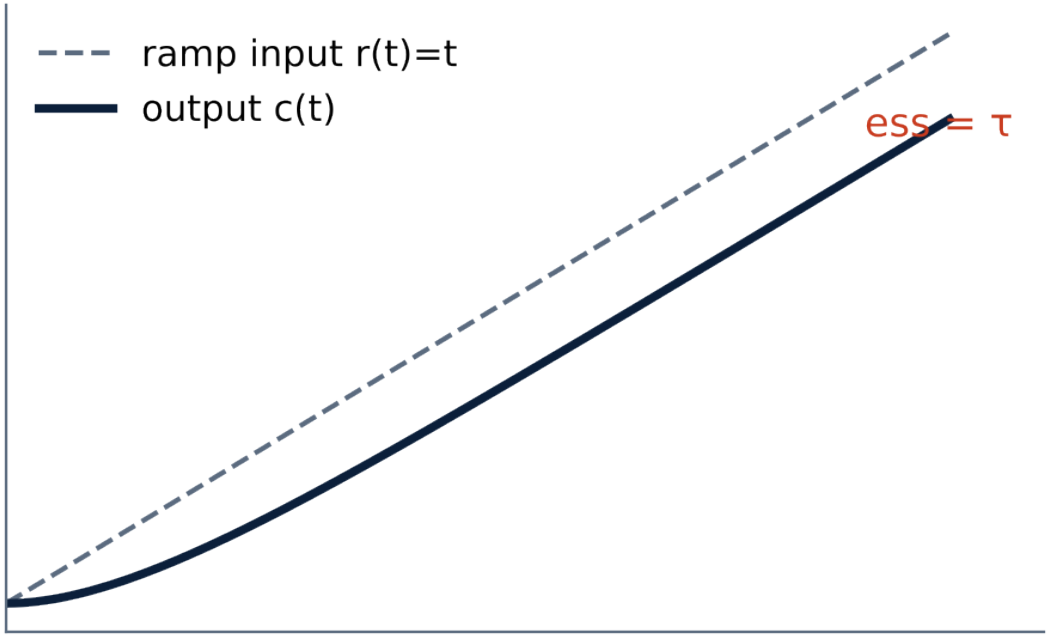
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First-order step response specifications

Specification	Definition	Value
Time constant T	Time to reach 63.2 %	T
Delay time td	Time to reach 50 %	0.693 T
Rise time tr	10 % to 90 %	2.2 T
Settling time ts (2 %)	Stay within ±2 %	4 T
Settling time ts (5 %)	Stay within ±5 %	3 T

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Ramp response and steady-state error



- c(t) = t − T + T e^(−t/T).
- Output follows the ramp with a constant lag.
- Steady-state error ess = T for a unit ramp.
- Error = r − c = T(1 − e^(−t/T)) → T.

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Numerical: time constant

Problem A system has G(s) = 2/(0.5s + 1). Find DC gain, time constant, rise time, 2 % settling time and final value for a unit step.

- Standard form K/(1 + sT): K = 2, T = 0.5 s
- tr = 2.2T = 1.1 s; ts(2 %) = 4T = 2 s
- Final value = K × 1 = 2
- c(t) = 2 (1 − e^(−2t))

Answer: K = 2, T = 0.5 s, tr = 1.1 s, ts = 2 s, c(∞) = 2

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Numerical: closed-loop first order

Problem A unity-feedback system has G(s) = 10/s. Find the closed-loop time constant and settling time.

- C/R = G/(1 + G) = (10/s)/(1 + 10/s) = 10/(s + 10) = 1/(0.1 s + 1)
- T = 0.1 s
- ts(2 %) = 4T

Answer: T = 0.1 s, ts = 0.4 s

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Key takeaways

- Zero order: c = K r (instantaneous)
- First order: G = 1/(1 + sT); step response 1 − e^(−t/T)
- tr = 2.2T, td = 0.69T, ts = 4T (2 %)
- Ramp: steady-state error = T

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Practice questions

Q1. Derive the unit step response of a first-order system.

Q2. T = 0.2 s: find tr, ts and the response at t = 0.4 s.

Q3. G(s) = 5/s with unity feedback: find T and ts.

Q4. Why is there no overshoot in a first-order system?